



Department of CSE IoT CSBT, VIT, Pune
Affiliated to
Savitribai Phule Pune University

Second Year (SY) Course Project (CP)

Project Report on

Project Title

Course Name:

Submitted by

- Student Name(s), Roll No.s, PRN
 - Department / Class
 - Guide Name
- Institute Name (e.g., Vishwakarma Institute of Technology)
- Academic Year (SEM-II 2025-2026)

Acknowledgment (new page)

- Brief gratitude to faculty, institute, team

Abstract (150–250 words) (new page)

- Problem statement
- Approach used (DSAOA concepts)
- Key results

Table of Contents (new page)

6. Introduction(new page)

- Background of the problem
- Motivation
- Real-world relevance

7. Problem Statement (new page)

- Clear and precise definition
- Input / Output specification
- Constraints

Objectives (new page)

- What the project aims to achieve

- Example:
 - Implement efficient data structures
 - Optimize time/space complexity

Literature Review (Short)

- Existing solutions / approaches
- Limitations of current methods

Methodology (New Page)

System Design / Approach

- Overall methodology
- Flowchart / Architecture diagram
- Modules description

Data Structures Used

Explain each with:

- Reason for selection
- Operations used

Example:

- Arrays / Linked List
- Stack / Queue
- Trees (BST, AVL, Heap)
- Graphs / Hashing

Algorithms Implemented

For each algorithm:

- Description
- Pseudocode
- Example

Implementation Details

- Programming language used
- Tools / IDE
- Code snippets (important parts only)

Complexity Analysis

- Time Complexity (Best / Average / Worst)
- Space Complexity

Present in table form: (Sample)

Operation	Time Complexity
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Insert	$O(\log n)$
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Search	$O(n)$
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Results and Tests (New Page)

Results and Output

- Screenshots

- Sample inputs & outputs
- Graphs (if performance comparison)

Testing

- Test cases (normal + edge cases)

Test Case	Input	Expected Output	Result
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Conclusion and Future Scope (New Page)

- **Applications** - Real-world use of the solution
- **Limitations** - Current drawbacks
- Summary of work done
- Learning outcomes
- **Future Scope** - Possible improvements

References (new Page)

- Books
- Research papers
- Websites

Appendix (Optional) (New Page)

- Full code
- Additional diagrams